

Basics 26: Ideal Ohm's Law and Flux Freezing

IV - Magnetohydrodynamics

The moving-fluid electric field, induction, Alfvén's theorem, topology, and nonideal breakdown

Textbook draft, version 1.0. The relation $\mathbf{E} + \mathbf{u} \times \mathbf{B} = 0$ is the kinematic core of ideal MHD. This chapter derives its interpretation, proves magnetic-flux conservation for a moving surface, and explains precisely how resistive and kinetic effects break ideal topology. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1	Learning objectives	2
2	Prerequisites and reading strategy	2
3	Electric field in the moving fluid frame	2
4	From generalized to ideal Ohm's law	3
5	Induction and magnetic-field stretching	3
6	Flux-freezing theorem	3
7	Physical interpretation and cautions	4
8	Breaking ideal flux freezing	4
9	Numerical flux freezing	4
10	Computational laboratory	4
11	Summary	4
	Symbols and acronyms used in this document	5

12 Exercises	5
13 References	5

1 Learning objectives

After completing this note, the reader should be able to:

- Interpret ideal Ohm's law in the fluid rest frame.
- Derive the ideal induction equation.
- Derive the field-to-density stretching relation.
- Prove flux freezing for a material surface.
- Distinguish flux conservation from constant field strength.
- Identify physical and numerical mechanisms that break flux freezing.

2 Prerequisites and reading strategy

- Basics 3, 12, 21, and 24.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Electric field in the moving fluid frame

At nonrelativistic speeds, the leading transformation to a frame moving with $\mathbf{u}$ is

$$\mathbf{E}' = \mathbf{E} + \mathbf{u} \times \mathbf{B}. \quad (3.1)$$

Ideal Ohm's law,

$$\boxed{\mathbf{E} + \mathbf{u} \times \mathbf{B} = 0}, \quad (3.2)$$

states that the electric field vanishes in the local bulk-fluid frame. It does not state that the laboratory electric field vanishes.

4 From generalized to ideal Ohm's law

Electron momentum balance gives schematically

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = \eta \mathbf{J} + \frac{\mathbf{J} \times \mathbf{B}}{en_e} - \frac{\nabla \cdot \mathbf{P}_e}{en_e} - \frac{m_e}{e} \frac{D_e \mathbf{u}_e}{Dt}. \quad (4.1)$$

Ideal MHD orders every term on the right small. The condition is therefore scale dependent. A plasma can be ideal globally while containing localized nonideal layers.

5 Induction and magnetic-field stretching

Faraday's law gives

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}). \quad (5.1)$$

Use

$$\nabla \times (\mathbf{u} \times \mathbf{B}) = \mathbf{u}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{u}) + (\mathbf{B} \cdot \nabla) \mathbf{u} - (\mathbf{u} \cdot \nabla) \mathbf{B}. \quad (5.2)$$

With $\nabla \cdot \mathbf{B} = 0$,

$$\frac{D\mathbf{B}}{Dt} = (\mathbf{B} \cdot \nabla) \mathbf{u} - \mathbf{B} \nabla \cdot \mathbf{u}. \quad (5.3)$$

Combining with $D\rho/Dt = -\rho \nabla \cdot \mathbf{u}$ gives

$$\boxed{\frac{D}{Dt} \left(\frac{\mathbf{B}}{\rho} \right) = \left(\frac{\mathbf{B}}{\rho} \cdot \nabla \right) \mathbf{u}}. \quad (5.4)$$

The field-to-mass ratio is stretched by velocity gradients.

6 Flux-freezing theorem

Let $S(t)$ move with the fluid. The transport theorem for magnetic flux gives

$$\frac{d}{dt} \int_{S(t)} \mathbf{B} \cdot d\mathbf{A} = \int_{S(t)} [\partial_t \mathbf{B} - \nabla \times (\mathbf{u} \times \mathbf{B})] \cdot d\mathbf{A}. \quad (6.1)$$

The bracket vanishes by the ideal induction equation:

$$\boxed{\frac{d\Phi_B}{dt} = 0}. \quad (6.2)$$

Equivalently, Faraday's law plus the motional electromotive force around a material loop gives zero net change.

7 Physical interpretation and cautions

If two ideal-fluid elements lie on the same magnetic field line initially, the mapping keeps them connected under smooth evolution. Field lines may be compressed, stretched, and folded, so B can change greatly even though material-surface flux is fixed. The statement is topological and Lagrangian; it is not the assertion that drawn field lines are physical strings embedded in matter.

8 Breaking ideal flux freezing

With resistivity,

$$\partial_t \mathbf{B} = \nabla \times (\mathbf{u} \times \mathbf{B}) + \eta_m \nabla^2 \mathbf{B} \quad (8.1)$$

for uniform magnetic diffusivity $\eta_m = \eta/\mu_0$. The magnetic Reynolds number

$$R_m = \frac{UL}{\eta_m} \quad (8.2)$$

compares advection with diffusion. Hall, pressure-tensor, inertia, finite-Larmor-radius, and turbulent effects can also break the bulk-fluid frozen-in relation, often with different species carrying distinct field-line velocities.

9 Numerical flux freezing

A discrete method may change topology because of truncation error even when the PDE is ideal. Excess numerical resistivity can damp Alfvén waves and alter eigenmode continua. Diagnostics include magnetic-energy decay, flux through advected loops, and convergence of reconnection rates toward zero in an ideal benchmark.

10 Computational laboratory

Advect a two-dimensional divergence-free magnetic field with a prescribed incompressible velocity. Track flux through a moving material loop and compare centered, upwind, and constrained-transport discretizations. Add controlled resistivity and recover the expected magnetic-diffusion scaling.

11 Summary

- Ideal Ohm’s law means the electric field vanishes in the bulk-fluid frame.
- It produces induction by advection, compression, and stretching of magnetic field.

- Magnetic flux through a material surface is conserved under smooth ideal evolution.
- Resistive, Hall, pressure, inertia, and kinetic effects can break the bulk frozen-in condition.
- Numerical diffusion can mimic nonideal physics and must be quantified.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$\mathbf{E}'$	moving-frame electric field	$\mathbf{E} + \mathbf{u} \times \mathbf{B}$.
Φ_B	magnetic flux	Surface integral of magnetic field.
η_m	magnetic diffusivity	η/μ_0 .
R_m	magnetic Reynolds number	Advection-to-diffusion ratio.
$S(t)$	material surface	Surface moving with the fluid.
D/Dt	material derivative	$\partial_t + \mathbf{u} \cdot \nabla$.

12 Exercises

1. Derive the expanded material form of the induction equation.
2. Combine mass and induction equations to obtain the evolution of $\mathbf{B}/\rho$.
3. Prove flux freezing using the moving-surface transport theorem.
4. Explain why magnetic-field strength can change under frozen flux.
5. Estimate the magnetic Reynolds number for a representative fusion-plasma scale.

13 References

References

- [1] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed., Cambridge University Press (2017).
- [2] J. D. Jackson, *Classical Electrodynamics*, 3rd ed., Wiley (1998).
- [3] F. F. Chen, *Introduction to Plasma Physics and Controlled Fusion*, 3rd ed., Springer (2016).

- [4] R. J. Goldston and P. H. Rutherford, *Introduction to Plasma Physics*, CRC Press (1995).
- [5] P. M. Bellan, *Fundamentals of Plasma Physics*, Cambridge University Press (2006).
- [6] J. P. Freidberg, *Ideal MHD*, Cambridge University Press (2014).
- [7] J. P. Goedbloed, R. Keppens, and S. Poedts, *Advanced Magnetohydrodynamics*, Cambridge University Press (2010).
- [8] H. Goedbloed and S. Poedts, *Principles of Magnetohydrodynamics*, Cambridge University Press (2004).